T96 fixed-input replay: target solid exclusion vs P0m hand layer vs no feature (pre-fixed criteria for solid delta 0 and delta 5)

Hand-layer rule	Remaining target voxels (band-excl.) request median	Median of 9 run medians	Per-request diff vs no feature: median (lower/equal/higher)	Remaining: visible / hidden (sum, 78 req.)	Finger-contact region remaining (median frac.)	Freed vs no feature (all requests)	Freed by GT class: apple / fruit / table / crate / robot / none	Fruit+table+crate share: all requests (window only)	Crit. 1: remaining < 5 %	Crit. 2: fruit+table+ crate <= 5 %
Original closed-loop record (replay noise floor)	0.1406	0.1431	-0.0001 (39/12/27)	39 / 3824	0.0000	471081	2149 / 205 / 6762 / 10947 / 72 / 113125	3.8 % (5.2 %)	-	-
No feature (surface extraction only)	0.1325	0.1371	reference	49 / 3796	0.0500	reference	-	-	-	-
P0m hand layer (hull fill + contact allowance 10 mm)	0.1153	0.1151	-0.0084 (51/27/0)	5 / 3165	0.0000	9037	6354 / 2015 / 0 / 0 / 204 / 464	22.3 % (35.2 %)	(not met)	(not met)
Target solid, delta 0 mm (tau 5 mm)	0.1299	0.1554	+0.0056 (28/10/40)	85 / 4056	0.0500	12454	12098 / 308 / 0 / 0 / 6 / 42	2.5 % (1.7 %)	not met	pass
Target solid, delta 5 mm (tau 5 mm)	0.1275	0.1554	+0.0014 (33/6/39)	85 / 3957	0.0500	16001	15169 / 356 / 0 / 0 / 18 / 458	2.2 % (1.5 %)	not met	pass
Target solid delta 0, table samples kept in d_N	0.1299	0.1554	+0.0056 (27/11/40)	85 / 4079	0.0500	12104	11819 / 255 / 0 / 0 / 6 / 24	2.1 % (1.7 %)	(not met)	(pass)
Target solid delta 0 + real-size palm spheres	0.1357	0.1411	+0.0081 (31/0/47)	89 / 3988	0.0513	0 (not comparable: grid origin differs 316/316 req., max 2.39 mm)	-	n/a	(not met)	n/a

Input: T87 fixed-input replay (B0 closed-loop k0 requests), 9 runs (hurdle 211 excluded), 435 requests, 78 grasp-window requests with a hand layer in every arm.

Definitions unchanged from outputs/2026-11-03_T87_.../재생_고정입력/README_definitions.md: grasp window = before GT grasp and palm-apple <= 150 mm; band = GT apple bottom + 6 mm excluded.

Visible = voxel holds a GT apple sample visible in the grasping-hand wrist camera at that request. Freed = no-feature <= 0 and arm > 0, MuJoCo GT class (nearest surface <= 8 mm).

Pre-fixed pass rule (T96 task 3.1): solid delta 0 and delta 5 each need crit. 1 and crit. 2. Values in parentheses are reported for other rows but are not criteria.

Real-size palm spheres row: hand-layer grid origin differs from the no-feature grid in every request (max 2.39 mm), so freed voxels are not computed (0).